

Machine Learning: Fundamental, Computational, and Conceptual Questions

Probability and Bayes' Theorem

Fundamental Questions

1. **What is a binary random variable? Provide examples.**

Answer: A binary random variable is a variable that can take one of two possible values, often representing the occurrence or non-occurrence of an event. It is associated with a probability that indicates the likelihood of the event occurring. Examples include:

- $A = \text{"I get heads in a coin toss"}$ (True/False).
- $A = \text{"It rains today"}$ (True/False).

The probability $P(A = \text{True})$ represents the fraction of possible worlds where A is true, and $P(A = \text{False}) = 1 - P(A = \text{True})$.

2. **State the axioms of probability.**

Answer: The axioms of probability are:

- (a) All probabilities are between 0 and 1: $0 \leq P(A) \leq 1$.
- (b) Valid propositions have probability 1, and unsatisfiable propositions have probability 0: $P(\text{True}) = 1$, $P(\text{False}) = 0$.
- (c) The probability of a disjunction is given by: $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$.

3. **What is the difference between joint and marginal probability?**

Answer:

- **Joint Probability:** The probability of two or more events occurring together, denoted as $P(A \wedge B)$ or $P(A, B)$. For example, $P(\text{alarm} \wedge \text{burglary})$ is the probability that both an alarm goes off and a burglary occurs.
- **Marginal Probability:** The probability of a single event, obtained by summing the joint probabilities over all possible values of other variables. For example, $P(\text{alarm}) = \sum_{b,e} P(\text{alarm} \wedge \text{burglary} = b \wedge \text{earthquake} = e)$.

4. **Define conditional probability and provide an example.**

Answer: Conditional probability, $P(A \mid B)$, is the probability of event A occurring given that event B is true. It represents the fraction of worlds where B is true in

which A is also true. For example, given a joint probability $P(\text{burglary} \wedge \text{alarm}) = 0.09$, $P(\text{alarm}) = 0.19$, and $P(\text{burglary}) = 0.1$, we compute:

$$P(\text{burglary} \mid \text{alarm}) = \frac{P(\text{burglary} \wedge \text{alarm})}{P(\text{alarm})} = \frac{0.09}{0.19} \approx 0.47.$$

5. **What is the difference between independence and conditional independence?**

Answer:

- **Independence:** Two events A and B are independent if $P(A \wedge B) = P(A)P(B)$, or equivalently, $P(A \mid B) = P(A)$. This means the occurrence of B does not affect the probability of A .
- **Conditional Independence:** Events A and B are conditionally independent given C if $P(A \wedge B \mid C) = P(A \mid C)P(B \mid C)$. This is a weaker condition, meaning A and B are independent when C is known.

Computational Questions

1. **Compute the joint probability $P(\text{alarm})$ given the joint distribution table below.**

Table:

	alarm, earthquake	alarm, $\sim$ earthquake	$\sim$ alarm, earthquake	$\sim$ alarm, $\sim$ earthquake
burglary	0.01	0.08	0.001	0.009
$\sim$ burglary	0.01	0.09	0.01	0.79

Answer: To compute $P(\text{alarm})$, sum the joint probabilities where the alarm is true:

$$\begin{aligned} P(\text{alarm}) &= P(\text{alarm} \wedge \text{burglary} \wedge \text{earthquake}) + P(\text{alarm} \wedge \text{burglary} \wedge \sim \text{earthquake}) + P(\text{alarm} \wedge \sim \text{burglary} \wedge \text{earthquake}) + P(\text{alarm} \wedge \sim \text{burglary} \wedge \sim \text{earthquake}) \\ &= 0.01 + 0.08 + 0.01 + 0.09 = 0.19. \end{aligned}$$

2. **Using Bayes' theorem, compute $P(\text{burglary} \mid \text{alarm})$ given $P(\text{burglary} \wedge \text{alarm}) = 0.09$, $P(\text{alarm}) = 0.19$, and $P(\text{burglary}) = 0.1$.**

Answer: Bayes' theorem states:

$$P(\text{burglary} \mid \text{alarm}) = \frac{P(\text{alarm} \mid \text{burglary})P(\text{burglary})}{P(\text{alarm})}.$$

First, compute $P(\text{alarm} \mid \text{burglary})$:

$$P(\text{alarm} \mid \text{burglary}) = \frac{P(\text{burglary} \wedge \text{alarm})}{P(\text{burglary})} = \frac{0.09}{0.1} = 0.9.$$

Then:

$$P(\text{burglary} \mid \text{alarm}) = \frac{P(\text{alarm} \mid \text{burglary})P(\text{burglary})}{P(\text{alarm})} = \frac{0.9 \times 0.1}{0.19} = \frac{0.09}{0.19} \approx 0.47.$$

3. In the Monty Hall problem, compute $P(A | B)$ where A is "the car is behind door 1" and B is "a goat is behind door 3" after the contestant picks door 1 and the host opens door 3.

Answer: Use Bayes' theorem:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

- $P(A) = \frac{1}{3}$ (the car is equally likely to be behind any door).
- $P(B | A)$: If the car is behind door 1 (A), the host opens door 2 or 3 (both have goats) with probability $\frac{1}{2}$. Thus, $P(B | A) = \frac{1}{2}$.
- $P(B)$: The probability that door 3 has a goat. Since door 3 has a goat in cases where the car is behind door 1 or 2:

$$P(B) = P(B | \text{car in 1})P(\text{car in 1}) + P(B | \text{car in 2})P(\text{car in 2}) + P(B | \text{car in 3})P(\text{car in 3}).$$

$$P(B) = \left(\frac{1}{2} \times \frac{1}{3}\right) + \left(1 \times \frac{1}{3}\right) + \left(0 \times \frac{1}{3}\right) = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}.$$

Thus:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}.$$

Therefore, the probability the car is behind door 1 is $\frac{1}{3}$, and switching to door 2 gives a $\frac{2}{3}$ chance of winning.

4. For a binomial distribution with $n = 17$ games and $k = 12$ wins, compute the maximum likelihood estimate (MLE) for the winning probability θ .

Answer: The MLE for a binomial distribution maximizes the likelihood:

$$P(D | \theta) = \binom{n}{k} \theta^k (1 - \theta)^{n-k}.$$

Taking the derivative of the log-likelihood and setting it to zero:

$$\frac{d}{d\theta} \left[\log \binom{n}{k} + k \log \theta + (n - k) \log(1 - \theta) \right] = \frac{k}{\theta} - \frac{n - k}{1 - \theta} = 0.$$

Solving:

$$\theta = \frac{k}{n} = \frac{12}{17} \approx 0.71.$$

Conceptual Questions

1. Explain the Monty Hall problem and why switching doors is advantageous.

Answer: In the Monty Hall problem, a contestant picks one of three doors, behind one of which is a car and behind the others are goats. After the contestant picks (say, door 1), the host, who knows what's behind the doors, opens another door (say, door 3) revealing a goat and asks if the contestant wants to switch to door 2. Initially, the probability of the car being behind door 1 is $\frac{1}{3}$. After the host reveals a goat behind door 3, the conditional probability $P(\text{car behind door 1} | \text{goat behind door 3})$ remains $\frac{1}{3}$ (as computed above). Thus, the probability the car is behind door 2 is $1 - \frac{1}{3} = \frac{2}{3}$. Switching is advantageous because it doubles the chance of winning from $\frac{1}{3}$ to $\frac{2}{3}$.

2. **Why is Bayes' theorem important in probabilistic machine learning?**

Answer: Bayes' theorem, $P(A | B) = \frac{P(B|A)P(A)}{P(B)}$, is crucial because it allows reasoning from observed evidence (data) to hypotheses (models or parameters). It updates prior beliefs ($P(A)$) with new data ($P(B | A)$) to compute posterior probabilities ($P(A | B)$). In machine learning, it's used to:

- Estimate model parameters (e.g., MAP estimation).
- Perform classification (e.g., Naive Bayes).
- Model uncertainty in probabilistic models.

3. **Compare Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP) estimation.**

Answer:

- **MLE:** Chooses the parameter θ that maximizes the likelihood of the observed data, $\theta^* = \arg \max_{\theta} P(D | \theta)$. It assumes no prior knowledge and relies solely on the data. For example, for $k = 12$ wins in $n = 17$ games, MLE gives $\theta = \frac{12}{17} \approx 0.71$.
- **MAP:** Chooses the parameter θ that maximizes the posterior probability, $\theta^* = \arg \max_{\theta} P(\theta | D) = \arg \max_{\theta} P(D | \theta)P(\theta)$. It incorporates prior knowledge via $P(\theta)$. For small datasets, MAP is more robust as it balances data and prior beliefs.

MLE is sensitive to small datasets, while MAP mitigates this by using priors.

4. **What is the significance of Laplace smoothing in Naive Bayes classifiers?**

Answer: Laplace smoothing addresses the issue of zero probabilities in Naive Bayes classifiers when a conditional probability (e.g., $P(A_i = a | C = c)$) is zero due to unseen attribute-class combinations in the training data. This causes the entire posterior probability to become zero, preventing classification. Laplace smoothing adjusts the probability as:

$$P(A_i = a | C = c) = \frac{N_{ac} + 1}{N_c + N_i},$$

where N_{ac} is the count of attribute value a in class c , N_c is the count of class c , and N_i is the number of unique values for attribute A_i . This ensures non-zero probabilities, enabling robust classification.

Naive Bayes Classifier

Fundamental Questions

1. **What is a Naive Bayes classifier?**

Answer: A Naive Bayes classifier is a probabilistic machine learning model used for classification. It assumes that features are conditionally independent given the class label, applying Bayes' theorem to compute the posterior probability of each class:

$$P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)},$$

where $X = (X_1, \dots, X_n)$ are features, and Y is the class. The "naive" assumption simplifies $P(X | Y)$ as:

$$P(X | Y) = \prod_{i=1}^n P(X_i | Y).$$

The class with the highest $P(Y | X)$ is chosen.

2. How does Naive Bayes handle continuous attributes?

Answer: For continuous attributes, Naive Bayes can:

- **Discretization:** Partition the continuous range into bins and treat the bin as a categorical value.
- **Probability Density Estimation:** Assume the attribute follows a distribution (e.g., normal). Estimate parameters (e.g., mean μ and variance σ^2) using MLE:

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i, \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

Compute the conditional probability using the probability density function, e.g., for a normal distribution:

$$P(X_i = x | Y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

3. What are the advantages of Naive Bayes classifiers?

Answer: Advantages include:

- Robust to noise, as conditional probabilities average out isolated noise points.
- Handles missing values by ignoring them during probability estimation.
- Robust to irrelevant attributes, as their conditional probabilities are uniformly distributed.
- Computationally efficient and easy to implement.
- Provides probabilistic outputs, offering insights into uncertainty.

Computational Questions

1. Using the dataset below, compute the Naive Bayes classification for a record $X = (\text{Refund} = \text{Yes}, \text{Status} = \text{Single}, \text{Income} = 80\text{K})$.

Dataset:

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

For Evade = Yes, $\mu = 90$, $\sigma^2 = 25$. For Evade = No, $\mu = 110$, $\sigma^2 = 2975$.

Answer: Compute $P(\text{Yes} | X)$ and $P(\text{No} | X)$:

- **Priors:** $P(\text{Yes}) = \frac{3}{10}$, $P(\text{No}) = \frac{7}{10}$.

- **Refund:**

$$P(\text{Refund} = \text{Yes} | \text{No}) = \frac{3}{7}, \quad P(\text{Refund} = \text{Yes} | \text{Yes}) = \frac{0}{3} = 0.$$

- **Marital Status:**

$$P(\text{Status} = \text{Single} | \text{No}) = \frac{2}{7}, \quad P(\text{Status} = \text{Single} | \text{Yes}) = \frac{2}{3}.$$

- **Income:** For Income = 80K, using normal distribution:

$$P(\text{Income} = 80 | \text{Yes}) = \frac{1}{\sqrt{2\pi \cdot 25}} e^{-\frac{(80-90)^2}{2 \cdot 25}} \approx 0.0108,$$

$$P(\text{Income} = 80 | \text{No}) = \frac{1}{\sqrt{2\pi \cdot 2975}} e^{-\frac{(80-110)^2}{2 \cdot 2975}} \approx 0.0062.$$

- **Posteriors:**

$$P(\text{Yes} | X) \propto P(\text{Yes}) \cdot P(\text{Refund} = \text{Yes} | \text{Yes}) \cdot P(\text{Status} = \text{Single} | \text{Yes}) \cdot P(\text{Income} = 80 | \text{Yes})$$

$$= \frac{3}{10} \cdot 0 \cdot \frac{2}{3} \cdot 0.0108 = 0.$$

$$P(\text{No} | X) \propto P(\text{No}) \cdot P(\text{Refund} = \text{Yes} | \text{No}) \cdot P(\text{Status} = \text{Single} | \text{No}) \cdot P(\text{Income} = 80 | \text{No})$$

$$= \frac{7}{10} \cdot \frac{3}{7} \cdot \frac{2}{7} \cdot 0.0062 \approx 0.0005.$$

Since $P(\text{No} | X) > P(\text{Yes} | X)$, classify as No. (Note: Laplace smoothing would avoid the zero probability for Yes.)

2. **Apply Laplace smoothing to compute $P(\text{Refund} = \text{Yes} | \text{Yes})$ and $P(\text{Marital Status} = \text{Married} | \text{Yes})$ for the above dataset.**

Answer: Laplace smoothing formula:

$$P(A_i = a | C = c) = \frac{N_{ac} + 1}{N_c + N_i}.$$

- **Refund:** For Evade = Yes, $N_{\text{Yes}} = 0$, $N_c = 3$, $N_i = 2$ (Yes/No).

$$P(\text{Refund} = \text{Yes} | \text{Yes}) = \frac{0 + 1}{3 + 2} = \frac{1}{5}.$$

- **Marital Status:** For Evade = Yes, $N_{\text{Married}} = 0$, $N_c = 3$, $N_i = 3$ (Single, Divorced, Married).

$$P(\text{Marital Status} = \text{Married} | \text{Yes}) = \frac{0 + 1}{3 + 3} = \frac{1}{6}.$$

Conceptual Questions

1. Why does the Naive Bayes classifier assume conditional independence?

Answer: The Naive Bayes classifier assumes conditional independence of features given the class label to simplify the computation of $P(X | Y)$. Without this assumption, computing the joint probability $P(X_1, \dots, X_n | Y)$ requires modeling all dependencies among features, which is computationally expensive and requires large amounts of data. The independence assumption allows:

$$P(X | Y) = \prod_{i=1}^n P(X_i | Y),$$

making the model scalable and effective, though it may not hold in practice if features are correlated.

2. How does Naive Bayes perform with missing attributes?

Answer: Naive Bayes handles missing attributes by ignoring them during probability calculations. For a record with missing attributes, only the available attributes contribute to the posterior probability:

$$P(Y | X_{\text{observed}}) \propto P(Y) \prod_{i \in \text{observed}} P(X_i | Y).$$

If all attributes are missing, it falls back to the prior probabilities $P(Y)$. This robustness makes Naive Bayes suitable for datasets with incomplete data.

3. Discuss the limitations of Naive Bayes classifiers.

Answer: Limitations include:

- **Independence Assumption:** The assumption of conditional independence may not hold, leading to degraded performance when features are correlated.
- **Zero Probabilities:** Without smoothing, unseen attribute-class combinations result in zero probabilities, preventing classification.
- **Biased Probability Estimates:** Naive Bayes probabilities are often biased, making it less reliable for precise probability outputs compared to models like logistic regression.
- **Limited Feature Incorporation:** It struggles to incorporate complex feature interactions or additional features beyond simple attributes (e.g., in text classification).

Techniques like Bayesian networks can address correlated features.

4. How is Naive Bayes applied to text classification?

Answer: In text classification, Naive Bayes treats each document as a bag of words, assuming words are conditionally independent given the class. For a document $d = (t_1, \dots, t_k)$ and class c :

$$P(c | d) \propto P(c) \prod_{i=1}^k P(t_i | c),$$

where $P(t_i | c)$ is the fraction of terms in class c that are t_i , often with Laplace smoothing:

$$P(t_i | c) = \frac{N_{ic} + 1}{N_c + T},$$

(N_{ic} : count of term t_i in class c , N_c : total terms in c , T : vocabulary size). It's effective for tasks like spam detection but struggles with complex features like word order.

Clustering

Fundamental Questions

1. **What is clustering, and how does it differ from supervised learning?**

Answer: Clustering is an unsupervised learning task that groups a set of objects into clusters such that objects within a cluster are similar to each other and dissimilar to those in other clusters. Unlike supervised learning, which uses labeled data to train a model for prediction, clustering works with unlabeled data to discover natural groupings. For example, clustering can group customers by purchasing behavior without prior labels, while supervised learning predicts customer categories based on labeled examples.

2. **What are the main types of clustering algorithms?**

Answer: The main types are:

- **Partitional Clustering:** Divides data into non-overlapping subsets (clusters), e.g., K-means.
- **Hierarchical Clustering:** Creates a nested set of clusters organized as a tree, either bottom-up (agglomerative) or top-down (divisive).

3. **Describe different types of clusters.**

Answer: Types of clusters include:

- **Well-Separated:** Each point is closer to every other point in its cluster than to any point outside it.
- **Prototype-Based:** Points are closer to their cluster's centroid (mean) or medoid (most representative point) than to other centroids.
- **Contiguity-Based:** Points are closer to at least one point in their cluster than to any point outside it.
- **Density-Based:** Clusters are dense regions of points separated by low-density regions, suitable for irregular shapes and outliers.

Computational Questions

1. **Outline the steps of the K-means clustering algorithm.**

Answer: The K-means algorithm is:

- (a) Select K points as initial centroids (often randomly).
- (b) Repeat:

- Assign each point to the cluster with the closest centroid (using a distance measure like Euclidean distance).
 - Compute the new centroid of each cluster as the mean of all points assigned to it.
- (c) Until the centroids do not change (or a stopping criterion is met, e.g., few points change clusters).
2. **Compute the Sum of Squared Error (SSE) for a K-means cluster with points $\{(1, 1), (2, 2), (3, 1)\}$ and centroid $(2, 1.33)$.**

Answer: SSE is the sum of squared Euclidean distances from each point to the centroid:

$$\text{SSE} = \sum_{x \in C_i} \text{dist}(x, c_i)^2.$$

For points $(1, 1)$, $(2, 2)$, $(3, 1)$ and centroid $(2, 1.33)$:

- Distance for $(1, 1)$: $\sqrt{(1-2)^2 + (1-1.33)^2} = \sqrt{1 + 0.1089} \approx 1.049$, squared: $1.049^2 \approx 1.1$.
- Distance for $(2, 2)$: $\sqrt{(2-2)^2 + (2-1.33)^2} = \sqrt{0 + 0.4489} \approx 0.67$, squared: $0.67^2 \approx 0.449$.
- Distance for $(3, 1)$: $\sqrt{(3-2)^2 + (1-1.33)^2} = \sqrt{1 + 0.1089} \approx 1.049$, squared: $1.049^2 \approx 1.1$.

$$\text{SSE} = 1.1 + 0.449 + 1.1 \approx 2.649.$$

3. Describe the K-means++ initialization algorithm.

Answer: K-means++ improves initialization to produce better clusters:

- Select the first centroid randomly from the data points.
- For $k - 1$ steps:
 - For each point x_i , compute the minimum squared distance to the existing centroids: $\min_j d^2(C_j, x_i)$.
 - Select a new centroid with probability proportional to $\frac{\min d^2(C_j, x_i)}{\sum_i \min d^2(C_j, x_i)}$.

This ensures centroids are spread out, reducing the chance of poor clustering.

Conceptual Questions

1. What are the limitations of the K-means algorithm?

Answer: Limitations include:

- **Sensitivity to Initialization:** Random centroid selection can lead to sub-optimal clusters; K-means++ mitigates this.
- **Fixed Number of Clusters:** The number of clusters K must be specified a priori.
- **Assumes Spherical Clusters:** Struggles with clusters of differing sizes, densities, or non-globular shapes.

- **Sensitive to Outliers:** Outliers can skew centroids, requiring preprocessing to remove them.
2. **How does K-means++ improve upon standard K-means?**
Answer: K-means++ improves initialization by selecting initial centroids that are spread out, reducing the likelihood of poor clustering outcomes. Unlike random selection in standard K-means, K-means++ chooses centroids with probability proportional to their squared distance from existing centroids, leading to lower SSE and more consistent results.
 3. **Explain how Gaussian Mixture Models (GMMs) differ from K-means.**
Answer:

- **Model Type:** K-means is a hard clustering algorithm assigning each point to one cluster based on the nearest centroid. GMM is a soft clustering probabilistic model, where each point has a probability of belonging to each cluster, modeled as a mixture of Gaussians:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k).$$

- **Cluster Shape:** K-means assumes spherical clusters (Euclidean distance). GMMs can model elliptical clusters via covariance matrices Σ_k .
- **Output:** K-means provides deterministic cluster assignments. GMMs provide posterior probabilities $p(z = k \mid \mathbf{x})$.
- **Algorithm:** K-means uses iterative assignment and centroid updates. GMMs use the EM algorithm to estimate parameters.

Expectation-Maximization (EM) Algorithm

Fundamental Questions

1. **What is the Expectation-Maximization (EM) algorithm?**
Answer: The EM algorithm is an iterative method for finding maximum likelihood estimates in models with latent variables, such as Gaussian Mixture Models. It alternates between:
 - **E-step:** Compute the posterior probabilities (responsibilities) of latent variables given the current parameters.
 - **M-step:** Update the model parameters to maximize the expected log-likelihood based on the responsibilities.

It's used when direct maximization of the likelihood is difficult due to unobserved variables.

2. **What is a Gaussian Mixture Model (GMM)?**
Answer: A GMM is a probabilistic model that represents a dataset as a mixture of K Gaussian distributions:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k),$$

where π_k are mixing coefficients ($\sum_k \pi_k = 1$, $\pi_k \geq 0$), μ_k is the mean, and Σ_k is the covariance matrix of the k -th Gaussian. GMMs are used for soft clustering and density estimation.

3. What is a latent variable in the context of GMMs?

Answer: In GMMs, the latent variable z represents the unobserved cluster assignment for each data point. It indicates which Gaussian component generated the point. Since z is not observed, it is marginalized out in the likelihood computation, and EM is used to infer its posterior probabilities.

Computational Questions

1. Describe the E-step and M-step in the EM algorithm for GMMs.

Answer:

- **E-step:** Compute the responsibility $r_k^{(i)}$ for each data point i and component k , which is the posterior probability:

$$r_k^{(i)} = p(z^{(i)} = k \mid \mathbf{x}^{(i)}; \theta) = \frac{\pi_k \mathcal{N}(\mathbf{x}^{(i)} \mid \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}^{(i)} \mid \mu_j, \Sigma_j)}.$$

- **M-step:** Update the parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$ to maximize the expected log-likelihood:

$$\begin{aligned} \pi_k &= \frac{1}{N} \sum_{i=1}^N r_k^{(i)}, \\ \mu_k &= \frac{\sum_{i=1}^N r_k^{(i)} \mathbf{x}^{(i)}}{\sum_{i=1}^N r_k^{(i)}}, \\ \Sigma_k &= \frac{1}{\sum_{i=1}^N r_k^{(i)}} \sum_{i=1}^N r_k^{(i)} (\mathbf{x}^{(i)} - \mu_k)(\mathbf{x}^{(i)} - \mu_k)^\top. \end{aligned}$$

2. For a GMM with two components, compute the responsibility $r_1^{(i)}$ for a data point $\mathbf{x}^{(i)} = 1$ in 1D, given $\pi_1 = 0.4$, $\pi_2 = 0.6$, $\mu_1 = 0$, $\mu_2 = 2$, $\sigma_1^2 = 1$, $\sigma_2^2 = 1$.

Answer: The responsibility is:

$$r_1^{(i)} = \frac{\pi_1 \mathcal{N}(x^{(i)} \mid \mu_1, \sigma_1^2)}{\pi_1 \mathcal{N}(x^{(i)} \mid \mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(x^{(i)} \mid \mu_2, \sigma_2^2)}.$$

Compute the Gaussian densities:

$$\mathcal{N}(x = 1 \mid \mu_1 = 0, \sigma_1^2 = 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(1-0)^2}{2 \cdot 1}} \approx 0.242,$$

$$\mathcal{N}(x = 1 \mid \mu_2 = 2, \sigma_2^2 = 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(1-2)^2}{2 \cdot 1}} \approx 0.242.$$

Weighted by mixing coefficients:

$$\pi_1 \mathcal{N}(1 \mid 0, 1) = 0.4 \times 0.242 \approx 0.0968,$$

$$\pi_2 \mathcal{N}(1 \mid 2, 1) = 0.6 \times 0.242 \approx 0.1452.$$

$$r_1^{(i)} = \frac{0.0968}{0.0968 + 0.1452} \approx 0.4.$$

Conceptual Questions

1. How is the EM algorithm related to K-means?

Answer: Both EM for GMMs and K-means are iterative clustering algorithms, but they differ in approach:

- **K-means:** Hard clustering with two steps:
 - Assignment: Assign each point to the nearest centroid.
 - Refitting: Update centroids as the mean of assigned points.
- **EM for GMMs:** Soft clustering with:
 - E-step: Compute posterior probabilities (responsibilities) for each point belonging to each Gaussian.
 - M-step: Update Gaussian parameters (mean, covariance, mixing coefficients) based on weighted data.
- **Relation:** K-means can be seen as a special case of EM where the covariance matrices are fixed, isotropic, and responsibilities are binary (0 or 1). EM generalizes K-means by allowing soft assignments and flexible cluster shapes.

2. Why is the EM algorithm suitable for GMMs?

Answer: The EM algorithm is suitable for GMMs because GMMs involve latent variables (cluster assignments z) that are not observed, making direct maximum likelihood estimation difficult. The log-likelihood includes a sum inside a logarithm:

$$\log p(\mathbf{X}; \theta) = \sum_{i=1}^N \log \sum_{k=1}^K p(\mathbf{x}^{(i)}, z^{(i)} = k; \theta),$$

which is hard to optimize. EM alternates between inferring the latent variables (E-step) and optimizing the parameters (M-step), iteratively improving the likelihood until convergence.

3. What are the advantages of GMMs over K-means?

Answer: Advantages of GMMs include:

- **Soft Clustering:** GMMs assign probabilities of membership to each cluster, allowing points to belong to multiple clusters with varying degrees.
- **Flexible Cluster Shapes:** GMMs model clusters with full covariance matrices, capturing elliptical and non-spherical shapes, unlike K-means' spherical assumption.
- **Probabilistic Framework:** GMMs provide a generative model, enabling density estimation and handling of uncertainty.
- **Robustness to Outliers:** Soft assignments make GMMs less sensitive to outliers compared to K-means' hard assignments.